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Presentation layer

The presentation layer is responsible for the UI of the application. It was built using jungle templates alongside bootstrap and crispy forms to improve the layout and make the pages easier to use. The users can register, login, send payments, request payments and view their transaction history through the web pages provided.

There's also separate pages which was created for the administrators. These pages allowed the admin to view the user accounts, view all transactions and create new admin accounts there's a navigation bar through the system which changes between whether you are normal user or an admin.

Business Logic Layer

The business logic is implemented through Django views, handling payments, payment request, validation, and access control. Transaction are processed inside the database transactions to ensure consistency and prevent partial updates, especially during payments. A RESTful service is used for currency conversion, ensuring accurate transfers between accounts with different currencies

Data Access Layer

The data access layer is built using Django models and SQLite, defining structured relationship between users, accounts, transactions and request, Each user has a single account, and all payments and request are stored with the relevant details for tracking and history. Django's ORM is used to safely manage database which made operations easier to manage and helped maintain data integrity.

Security Layer

The security layer ensures that users and administrators can only access authorised parts of the application. Django's built-in authentication system was used to support registration, login, and logout functionality.

Access control was added so that only logged-in users can use the payment system, while administrator pages are restricted to admin users only. HTTPS support was also implemented using a self-signed SSL certificate to provide encrypted communication between the browser and the application.

The system also benefits from Django's built-in protection against common web vulnerabilities such as Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), SQL injection, and clickjacking attacks.

Web Services

A RESTful web service was created to handle currency conversion between GBP, USD, and EUR. The service uses predefined exchange rates and responds to GET requests using JSON format.

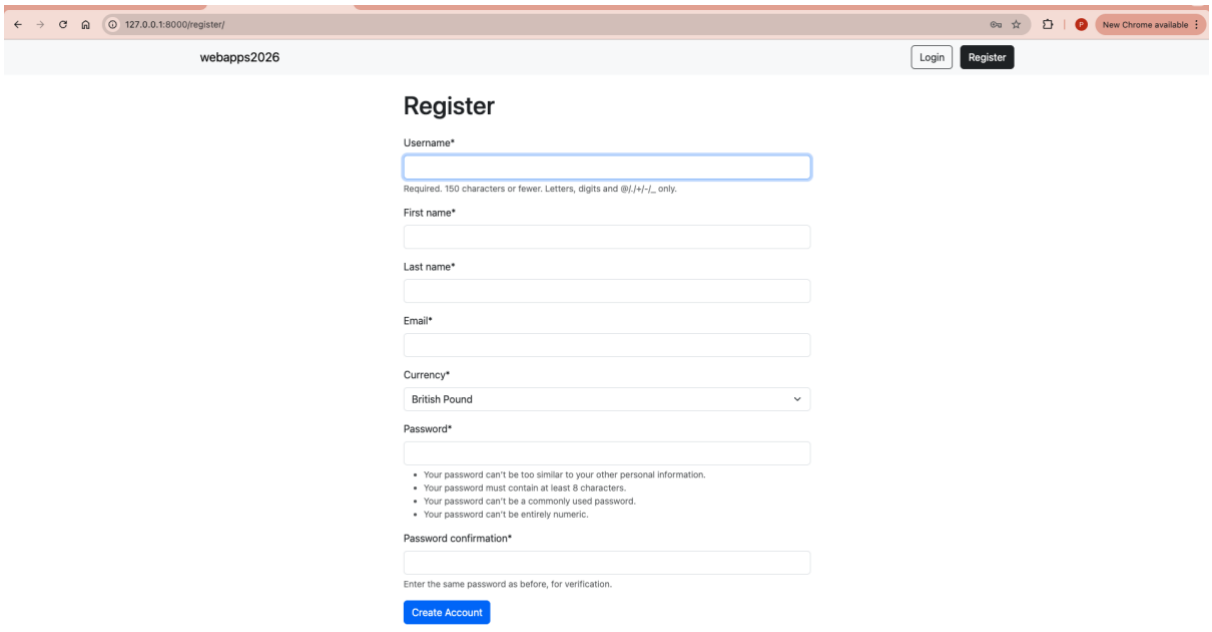
The service is accessed automatically whenever users transfer money between accounts with different currencies. It returns the conversion rate and converted amount, which are then used to update account balances correctly. This separates the currency conversion into its own web service keeps the application more organised and makes the conversion logic easier to manage.

User Manual

This section provides a walkthrough of the main features of the system, showing how users and administrators can interact with the application. Screenshots are included to illustrate each step.

User Registration

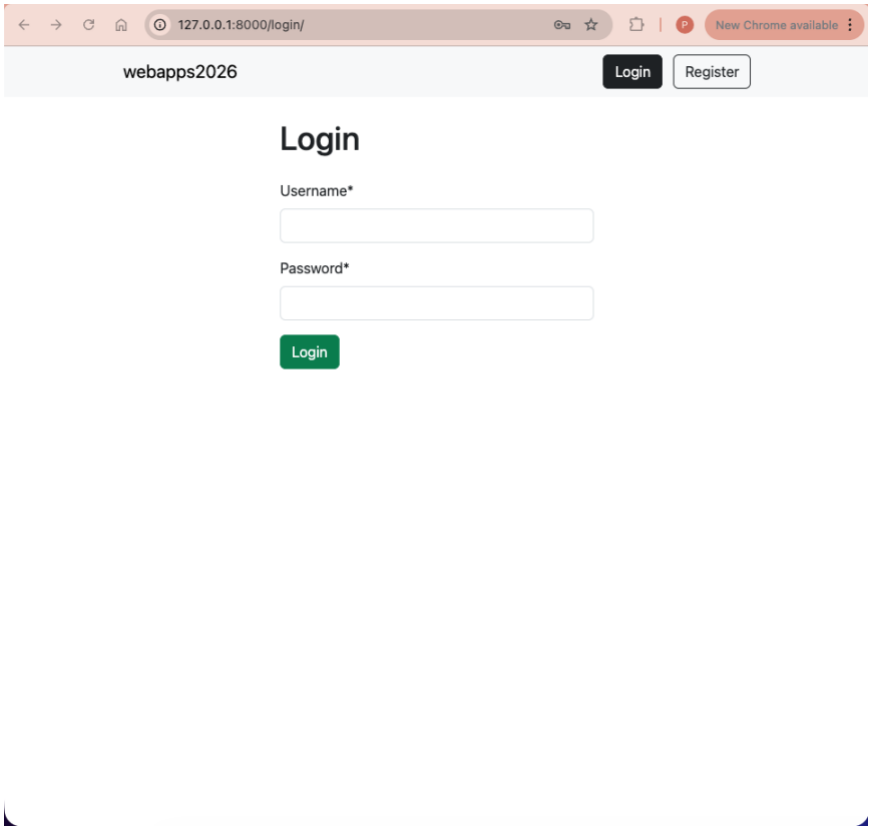
New user Can create an account through the registration page by entering their username, personal details, e-mail address, passwords and currency. Once the user has registered an account will be automatically created with a balance converted in the selected currency.



/Register/

User Login

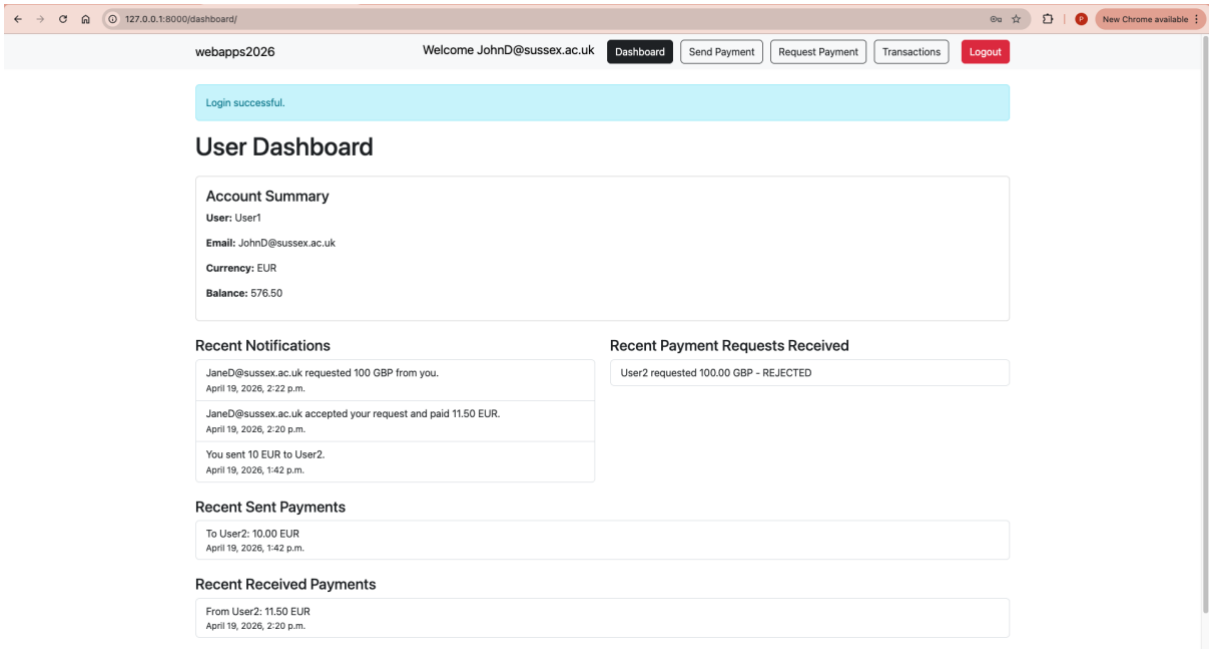
Once the user has registered, they can log into a system using their username and password. After authentication users will be redirected to their home dashboard



/Login/

User Dashboard

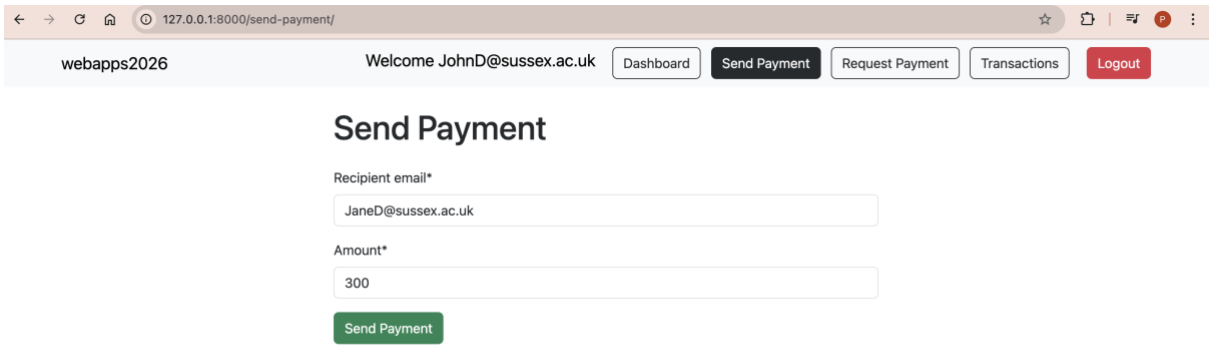
The dashboard provides an overview of the user account including current balanced, amount currency, recent notifications, recent transactions, recent payment request.



/Dashboard/

Sending Payments

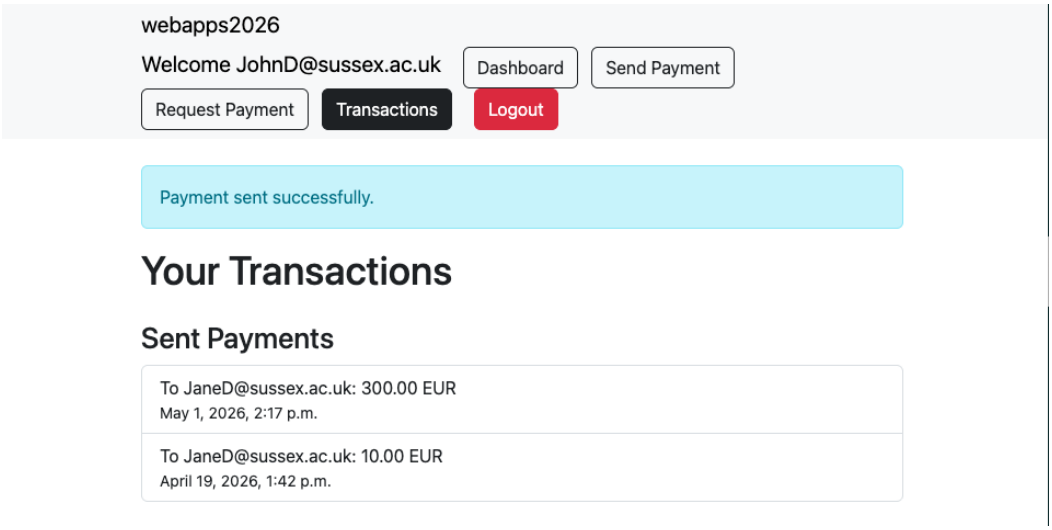
Users can send money to other registered users enter in the other users e-mail address and the payment amount. The system then validates that the other user exist, the sender has enough balance and is not transferring money to themselves.



/send-payment/

Successful payment

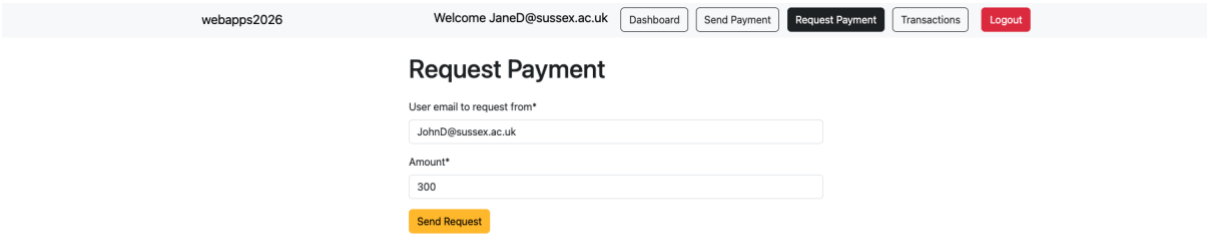
Once the payment has been successfully completed the system then displays a confirmation message and updates both account balances.

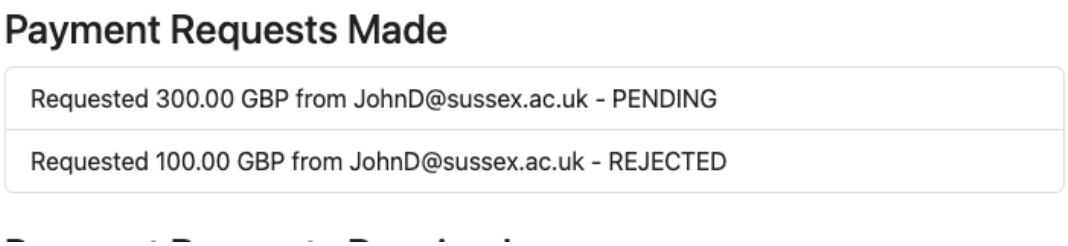


Confirmation message showing a successful payment transaction.

Requesting Payments

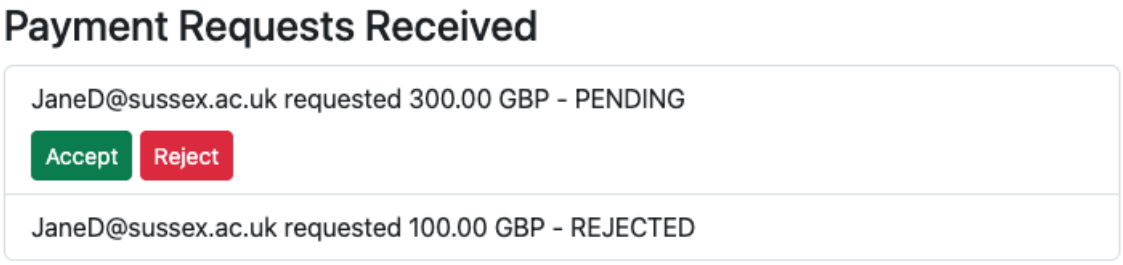
Users can also request money from another registered user by entering the other users e-mail address and the requested amount. Once the request has been sent the request remains pending until the recipient of the payment either accepts or rejects





Accepting and Rejecting Requests

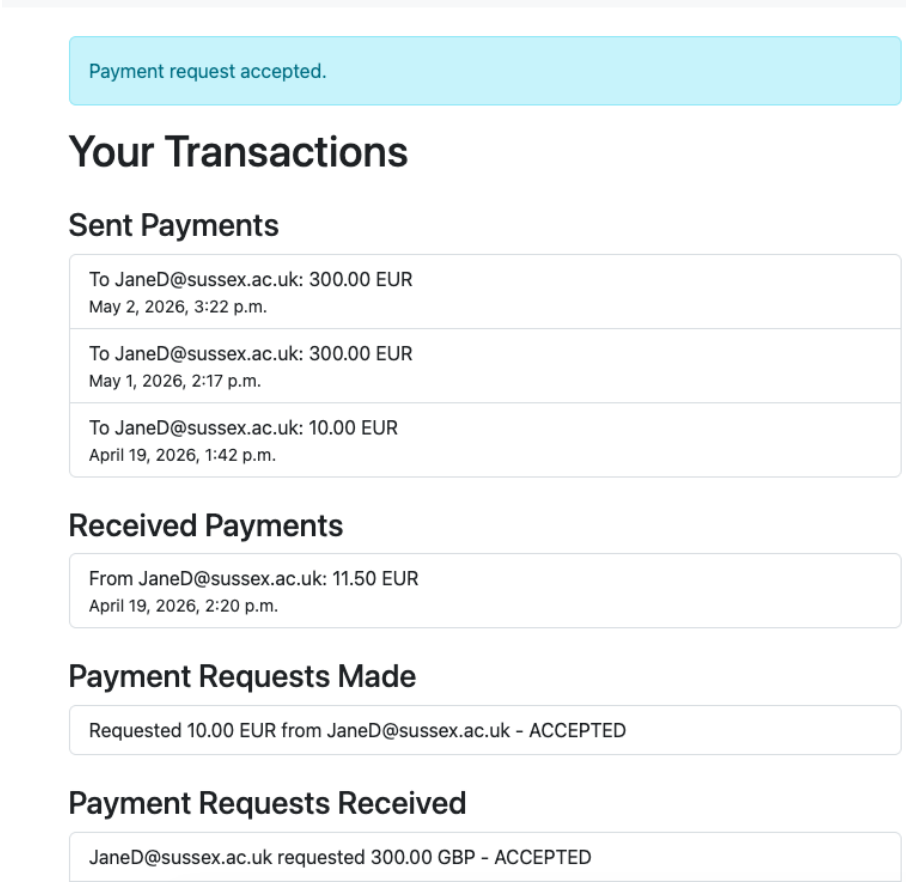
Once the recipient Caesar incoming payment they can either accept the request, which automatically transfers the funds or reject the requests, which updates the request status without transferring money.



/request-payment/

Transactions

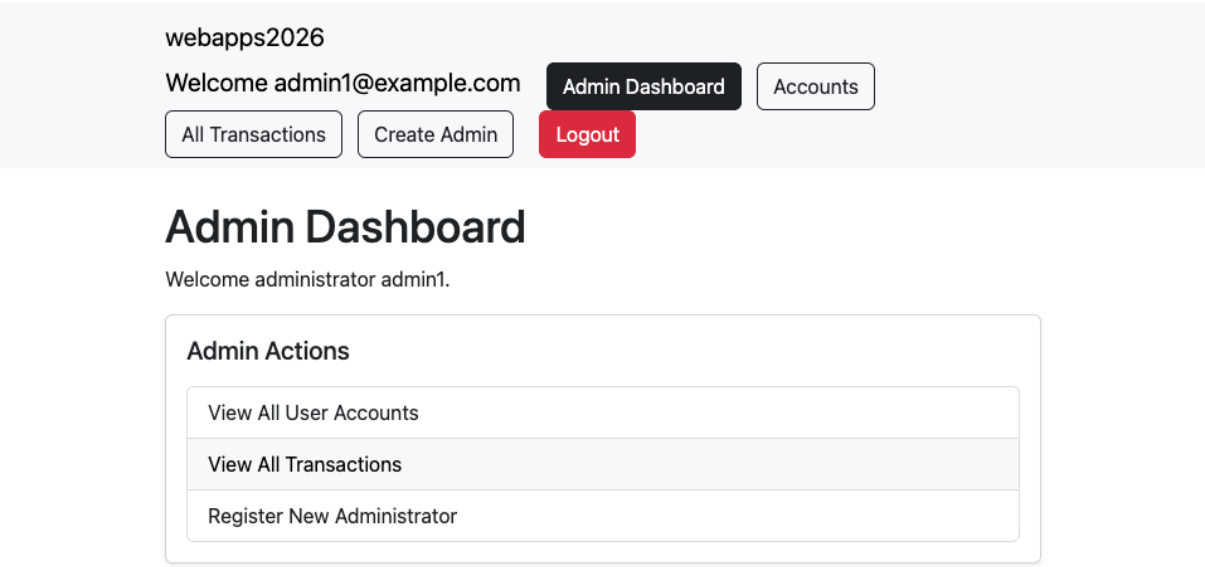
The transaction page shows a complete history of sent payments, received payments, request made and requests received Which allows the users to track all activity associated to their account



/Transactions/

Admin Dashboard

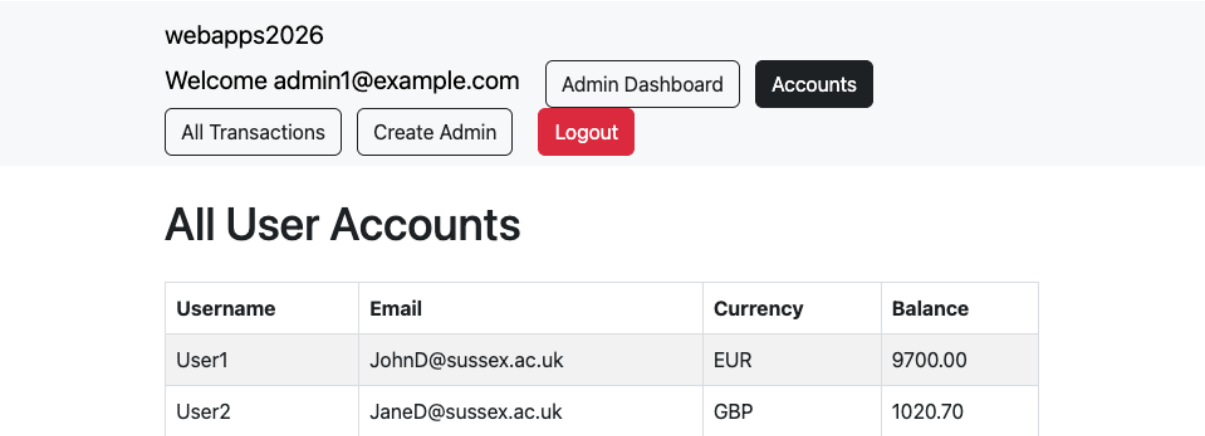
The administrators have access to a separate dashboard which allows them to view all user accounts view all system transactions and create additional administrator accounts.



/admin-dashboard/

Viewing User Accounts

Administrators can see a complete list of all the registered users and their account balances



/staff-accounts/

Admin Transactions

Admins can view all the transactions and payment request stored in the system

Admin Transactions

Completed Payments

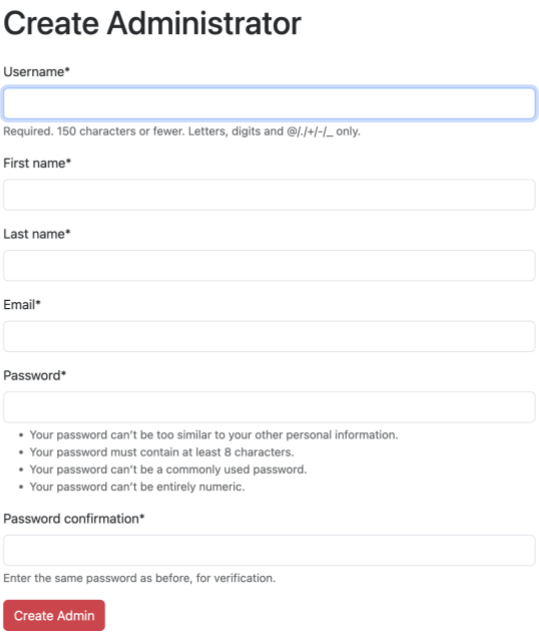
Sender	Recipient	Amount Sent	Amount Received	Rate	Type	Date
JohnD@sussex.ac.uk	JaneD@sussex.ac.uk	300.00 EUR	261.00 GBP	0.870000	REQUEST	May 2, 2026, 3:22 p.m.
JohnD@sussex.ac.uk	JaneD@sussex.ac.uk	300.00 EUR	261.00 GBP	0.870000	DIRECT	May 1, 2026, 2:17 p.m.
JaneD@sussex.ac.uk	JohnD@sussex.ac.uk	10.00 GBP	11.50 EUR	1.150000	REQUEST	April 19, 2026, 2:20 p.m.
JohnD@sussex.ac.uk	JaneD@sussex.ac.uk	10.00 EUR	8.70 GBP	0.870000	DIRECT	April 19, 2026, 1:42 p.m.

Payment Requests

Requester	Target User	Amount	Currency	Status	Created	Responded
JaneD@sussex.ac.uk	JohnD@sussex.ac.uk	300.00	GBP	ACCEPTED	May 2, 2026, 2:54 p.m.	May 2, 2026, 3:22 p.m.
JaneD@sussex.ac.uk	JohnD@sussex.ac.uk	100.00	GBP	REJECTED	April 19, 2026, 2:22 p.m.	April 19, 2026, 2:23 p.m.
JohnD@sussex.ac.uk	JaneD@sussex.ac.uk	10.00	EUR	ACCEPTED	April 19, 2026, 2:17 p.m.	April 19, 2026, 2:20 p.m.

Create Admin

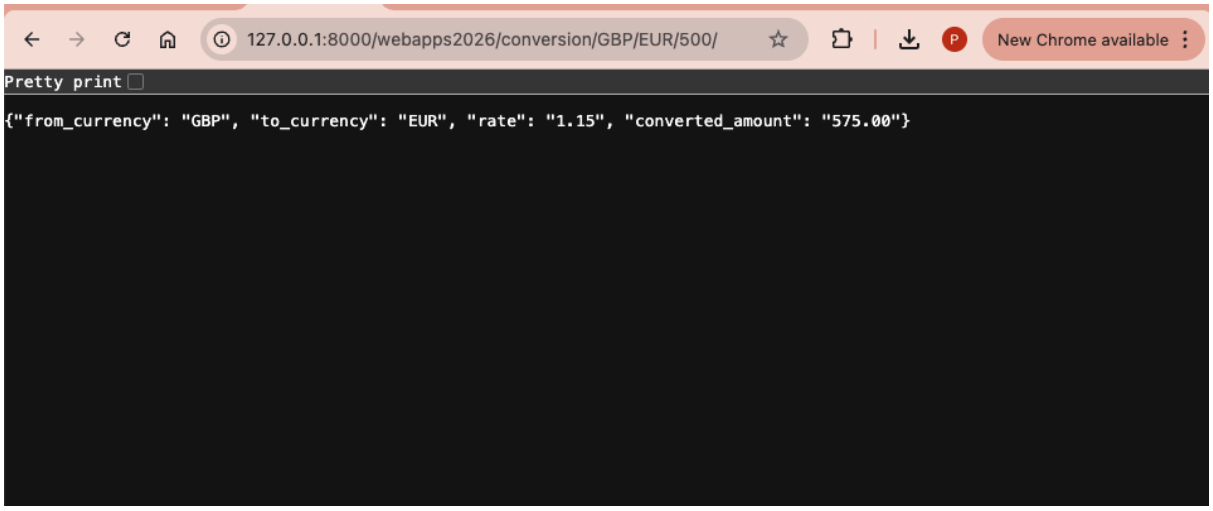
Admins can register more admin users through the create admin page



Create Admin

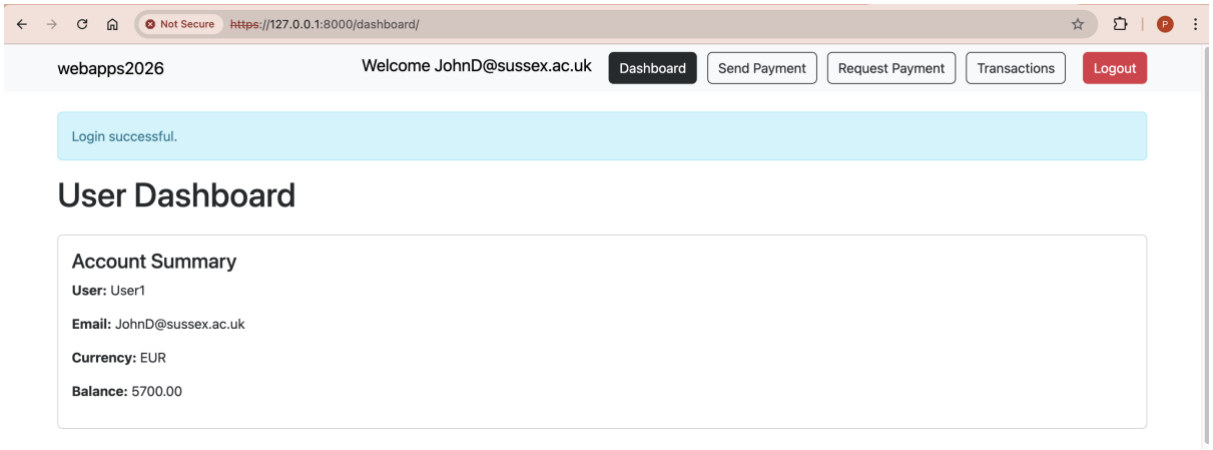
REST API Currency Conversion

The system includes a RESTful API endpoint used for currency conversion. The API returns JSON responses containing The original amount, converted amount and exchange rate.



HTTPS

This was configured using Django Extensions and a self-signed SSL certificate to demonstrate encrypted communication between the client and server.





Deployment on AWS

The application was deployed on AWS EC2 virtual machine to make it accessible online. An Ubuntu server instance was created through the AWS Academy and the jungle project was uploaded to the server using SSH commands



